Sourab Sridhar

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WORK EXPERIENCE

Qualcomm Inc **Stockholm, Sweden**

Staff Software Engineer *Dec 2025 - Present*

- Led architecture improvements for end-to-end autonomous driving Vision-Language-Action (VLA) foundation models, engineering specialized components to resolve critical gaps in temporal understanding.
- Advanced VLA model capabilities to tackle unpredictable decision-making in safety-critical environments, significantly improving system responsiveness and traffic light compliance at complex intersections.

Senior Software Engineer *Dec 2023 - Nov 2025*

- Trained and deployed DETR-based detectors and Kalman filter trackers using PyTorch and MMDetection3D, improving detection by 15+ mAP points and enabling auto-labeling of 330K+ sequences for Static 3DOD missions across varied ODDs and data from 60+ countries.
- Built scalable cloud-based auto-labeling pipelines with Argo Workflows, automating 690K+ sequences with >95% yield, saving USD 15M+. Used Weights & Biases for experiment tracking and achieved 36% better results than manual annotations, outperforming legacy systems by 40+ mAP points.
- Delivered validation data for multiple 3DOD releases on time through collaboration with 3+ departments; submitted 3 Invention Disclosure Forms (IDFs) to strengthen Qualcomm's IP portfolio and updated internal repositories for production readiness. Supported customer demos to drive adoption of automation pipelines.

Software Engineer *Mar 2023 - Nov 2023*

- Developed and deployed lidar-only auto-labeling models for static 3D objects using a LargeKernel3D encoder and TransFusion based detection head, achieving >90% mAP at 50m.
- Led end-to-end development from data preprocessing to model training using PyTorch and MMDetection3D, integrating open-source and internal datasets to ensure robustness.
- Scaled auto-labeling to 40K+ sequences via cloud deployment with <5 min inference per sequence and showcased results at CES 2024 with strong stakeholder validation.

Semcon Sweden AB **Gothenburg, Sweden**

Deep Learning Engineer *Jun 2021 - Mar 2023*

- Led the setup and debugging of the first Static 3DOD data pipeline, resolving complex lidar aggregation issues (e.g., INS drift) to generate high-quality point clouds for model training.
- Built scalable on-prem pipelines and an annotation validation framework with sanity checkers, improving data quality across 5+ missions and supporting timely delivery for 10+ annotation projects.
- Contributed to model development and evaluation using PyTorch and MMDetection3D, bridging data engineering and deep learning for static object detection.

Master Thesis Student *Jan 2021 - Nov 2021*

- Developed a novel real-time pedestrian intent prediction model using Graph Neural Networks (GNNs) and PyTorch Geometric, modeling scene interactions as graphs.
- Achieved 52% accuracy with detailed evaluation (precision, recall, F1-score) and 6x faster inference than state-of-the-art models, running at 6 FPS.
- Benchmarked 5+ state-of-the-art models and trained the novel architecture on JAAD and PIE datasets to ensure robustness and generalizability using PyTorch.

Robert Bosch GmbH **Bangalore, India and Abstatt, Germany**

Senior Software Engineer *Apr 2018 - Jun 2019*

- Provided support for requirement engineering and developed test strategies for device drivers and ECU hardware verification software, initiating development activities for the next generation Bosch electrical braking system.
- Improved software variant management for safety-critical supply modules across all Bosch electrical braking system products, reducing software maintenance efforts by 50% for 30+ projects globally.
- Resolved over 25 production issues worldwide with an average resolution time of <3 days, lowering the defect rate to 5 ppm.

Product Software Engineer *Mar 2017 - Apr 2018*

- Collaborated with over 60 stakeholders globally to establish the first "Redundant Supply Concept" for Bosch electrical braking system products, leading to the development of two active safety products compatible with autonomous driving level 3.
- Developed device drivers and conducted hardware and software testing for safety-critical redundant supply modules in all Bosch electrical braking system products, supporting the series production of 5+ autonomous driving level 3 compatible active safety projects worldwide.

Associate Software Engineer *Jul 2015 - Mar 2017*

- Developed device drivers for safety-critical supply modules across all Bosch electrical braking system products, achieving ASIL-D compliance in 30+ projects worldwide.
- Created and implemented simulation models and tests for over 60% of Bosch electrical braking system-specific device drivers, increasing software testing coverage by more than 500% in 30+ projects globally.
- Maintained ASPICE process compliance in all personal development tasks, contributing to ASPICE level 1 assessment for all Bosch electrical braking system products.

EDUCATION

Chalmers University of Technology **Gothenburg, Sweden**

MSc. Systems, Controls and Mechatronics *Graduation Date: Nov 2021*

Visveswarya Technological University **Bangalore, India**

BE. Electrical and Electronics Engineering *Graduation Date: Jun 2015*

TECHNICAL SKILLS

- **Skills:** Deep Learning (CNNs, Transformers, VLMs), Computer Vision (2D/3D Object Detection, Segmentation, Tracking), Sensor Fusion (LiDAR, Camera)
- **Languages:** Python, C/C++
- **Libraries:** PyTorch, NumPy, OpenCV, Open3D, Pandas, MMDetection3D
- **Tools:** Git, Weights and Biases, Argo, AWS, IBM ALM, Rational Doors, Jira, UML

PUBLICATIONS AND PATENTS

- HAN, S.S., Cai, H., Wang, H., ZHU, Y., SHI, Y., PORIKLI, F.M., **SRIDHAR, S.B**. and Yogamani, S.K., Qualcomm Inc, 2026. *Three-dimensional (3D) object detection based on multiple two-dimensional (2D) views*. U.S. Patent 12,614,395.
- **SRIDHAR, S.B**., Narayanan, V., Kumar, V.R., Yogamani, S.K. and RAVI, K.B., Qualcomm Inc, 2025. *Traffic object recognition systems and methods*. U.S. Patent Application 18/605,264.
- **Bapu Sridhar, S**. and Moosakhanian, A., 2020. Pedestrian Intent Prediction Using Deep Machine Learning.